MAIA - Minimal Alchemical Intelligence Agent

Welcome to MAIA, a universal cognitive bus that runs on any terminal with a Python 3 environment. This package provides a self-contained, portable AI agent capable of reasoning, learning, and adapting.

This document provides instructions for installing and running MAIA on your system, including mobile devices using Termux.

Installation

Prerequisites

- Python 3.x
- `git` (for cloning the repository)

Desktop (Linux/Mac/Windows)

1. **Clone the repository:**

   ```bash
   git clone <repository_url>
   cd MAIA_Package
   ```

2. **Run MAIA:**

   ```bash
   python3 maia_core.py
   ```

Mobile (Android with Termux)

1. **Install Termux:**

   Install Termux from the Google Play Store or F-Droid.

2. **Update and install dependencies:**

   Open Termux and run the following commands:

```bash
pkg update && pkg upgrade -y
pkg install python git -y
```

3. **Create a directory and clone the repository:**

```bash
mkdir -p /storage/emulated/0/Infinite_Scroll/maia
cd /storage/emulated/0/Infinite_Scroll/maia
git clone <repository_url>
cd MAIA_Package
```

4. **Run MAIA:**

```bash
python maia_core.py
```

Usage

Once MAIA is running, you can interact with it through the command-line interface. Here are the available commands:

- **<query>** : Enter any vague query or task for MAIA to process. MAIA will decompose it into sub-goals, reason about them, and provide a summary of its process.
- **report** : Generate a reflection report on the current session, including memory statistics and recent decisions.
- **memory** : Display the current state of MAIA's memory, including the number of beliefs, goals, contradictions, and decisions.
- **operator <command>** : Access operator controls for manual intervention. Available commands are:
  - `override` : Pause MAIA's processing to prevent it from automatically handling new queries.
  - `resume` : Resume normal processing.
  - `status` : Check the current session status and memory statistics.
  - `save` : Manually save the current memory state to disk.
- **exit** : Shut down the MAIA node. This will also save the current memory state.

Troubleshooting

- **`ModuleNotFoundError: No module named '...'`** : This indicates a missing Python package. While MAIA aims for minimal dependencies, you can install any required packages using `pip install <package_name>`.
- **Permission errors on Android**: If you encounter permission errors when creating directories or files, ensure Termux has storage permissions. You can grant this by running `termux-setup-storage` and then accessing `/storage/emulated/0/`.
- **Memory file issues**: If you see warnings about memory loading, check the `maia_memory` directory for corrupted JSON files. You can safely delete the contents of this directory to start with a fresh memory, but you will lose all previous beliefs and decisions.